

IEEE PES Task Force on Benchmark Systems for Stability Controls

Report on the Benchmark System #2 7-bus, 5-machine Equivalent Brazilian System

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1. Introduction	2
2. Power Flow	2
3. Dynamic Simulation Models.....	6
3.1 Synchronous Machines	6
3.2 Excitation Systems	8
3.3 Power System Stabilizers	8
4. Simulation Results.....	10
4.1 50 Mvar Reactor at Ivaiporã Bus	11
4.2 Step at Itaipu Voltage Reference.....	16
4.3 Modal Analysis	21
4.3.1 No PSS	21
4.3.2 PSS at Itaipu	23
4.3.3 PSS at All Machines.....	24
5. References	25

1. Introduction

This report presents the data setup and nonlinear stability study carried over with the 5-generators equivalent system of the Itaipu 60 Hz transmission system proposed in [1] using the Siemens PTI's PSS[®]E software [2]. The main objectives of this report are to document the data setup and to provide some validation of such data, comparing (to the extent possible) the results obtained with a time-domain nonlinear simulation with the eigenvalue analysis shown in [1].

2. Power Flow

The power flow solution is shown in the one line diagram in Figure 1. The bus data, including the power flow solution, are shown in Table 1.

The load data is given in Table 2 and negative loads (active or reactive parts) imply current being injected into the grid. These loads are represented, in the dynamic simulation, with a constant current characteristic for the active power and a constant admittance characteristic for the reactive power (100%I, 100%Z for P and Q, respectively).

The shunt data is shown in Table 3. The values for these capacitors are shown at nominal voltage (1.0 pu voltage).

The transmission line data is shown in Table 4. The data is provided in per unit considering a system MVA base of 1,000 MVA. The lines are represented by series (RL) impedances. The capacitive chargings of the lines have been equivalenced with line and bus reactors and are accounted for in the shunt data shown in Table 3.

The transformer data is given in per unit on system MVA base (1,000 MVA) in Table 5.

The machine data is shown in Table 6. The maximum active power output is defined to be equal to the machine MVA base and the reactive power output is defined based on the power factor of 0.8. The generators step-up transformers (GSU) are not represented in the case.

The complete power flow solution report for this system is given in Table 7 (PSSE activity LOUT).

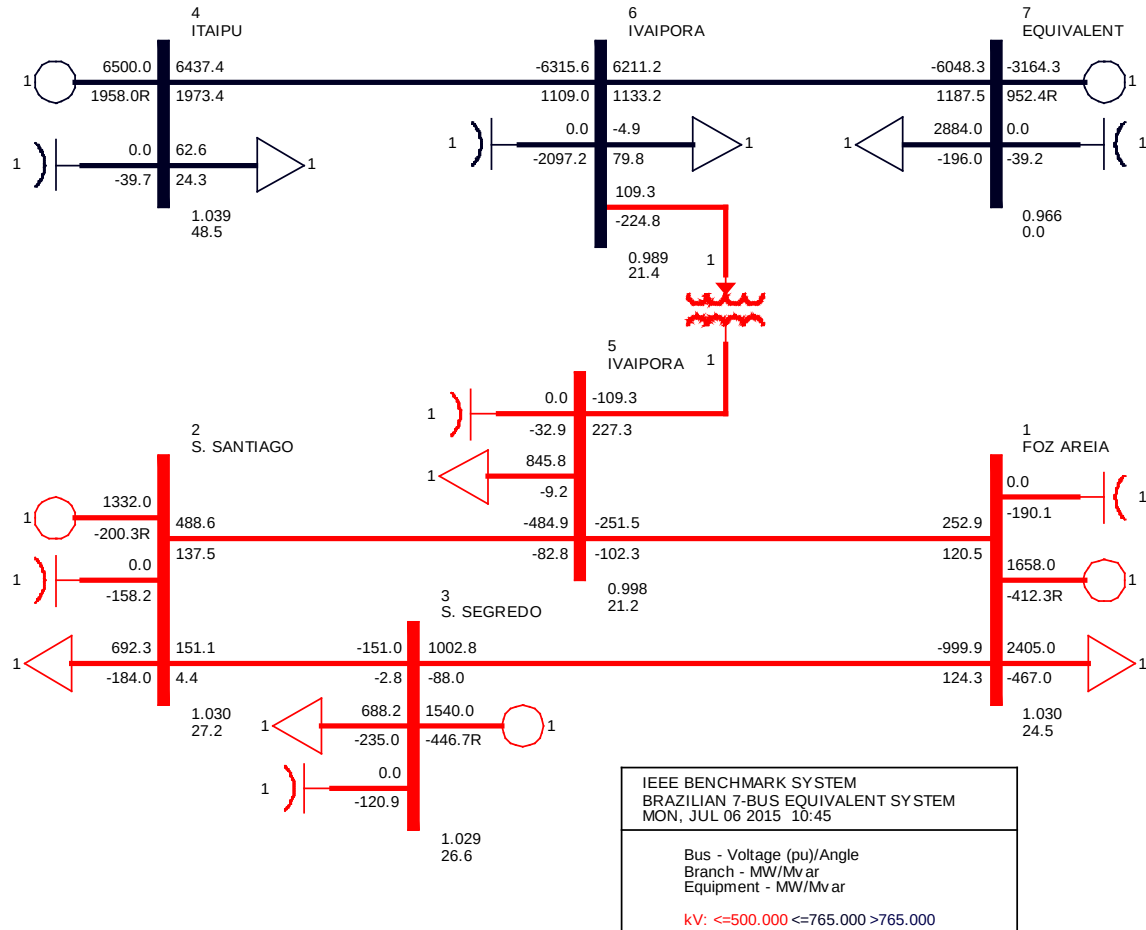


Figure 1: Power Flow Solution

Table 1: Bus Data and Power Flow Solution

Bus Number	Bus Name	Base kV	Bus type	Voltage (pu)	Angle (deg)
1	FOZ AREIA	500.0	PV	1.0300	24.53
2	S. SANTIAGO	500.0	PV	1.0300	27.22
3	S. SEGREDO	500.0	PV	1.0290	26.60
4	ITAIPU	765.0	PV	1.0390	48.45
5	IVAIPORA	500.0	PQ	0.9984	21.20
6	IVAIPORA	765.0	PQ	0.9895	21.45
7	EQUIVALENT	765.0	swing	0.9660	0.00

Table 2: Load Data

Bus Number	P (MW)	Q (Mvar)
1	2405.0	-467.0
2	692.3	-184.0
3	688.2	-235.0
4	62.6	24.3
5	845.8	-9.2
6	-4.9	79.8
7	2884.0	-196.0

Table 3: Shunt Data

Bus Number	Bus Name	Base kV	G-Shunt (MW)	B-Shunt (Mvar)
1	FOZ AREIA	500.0	0.0	179.2
2	S. SANTIAGO	500.0	0.0	149.1
3	S. SEGREDO	500.0	0.0	114.2
4	ITAIPU	765.0	0.0	36.8
5	IVAIPORA	500.0	0.0	33.0
6	IVAIPORA	765.0	0.0	2142.0
7	EQUIVALENT	765.0	0.0	42.0

Table 4: Transmission Line Data (1,000 MVA Base)

From Bus	To Bus	R (pu)	X (pu)	Charging (pu)
1	3	0.003	0.038	0.0
1	5	0.019	0.245	0.0
2	3	0.005	0.076	0.0
2	5	0.015	0.225	0.0
4	6	0.0029	0.0734	0.0
6	7	0.004	0.057	0.0

Table 5: Transformer Data (1,000 MVA Base)

From Bus	To Bus	R (pu)	X (pu)	tap (pu)
5	6	0.0	0.039	1.0

Table 6: Power Flow Generator Data

Bus Number	VSched (pu)	Pgen (MW)	Qgen (Mvar)	Pmax (MW)	Qmax (Mvar)	Qmin (Mvar)	Mbase (MVA)
1	1.030	1658.0	-412.3	1900	1140	-1140	1900
2	1.030	1332.0	-200.3	1400	840	-840	1400
3	1.029	1540.0	-446.7	1944	1166	-1166	1944
4	1.039	6500.0	1957.9	6633	3980	-3980	6633
7	0.966	-3164.3	952.3	3000	3600	-3600	6000

Table 7: PSS®E Power Flow Results

PTI INTERACTIVE POWER SYSTEM SIMULATOR--PSS(R)E																	WED, OCT 02 2013 10:17	
PTI INTERACTIVE POWER SYSTEM SIMULATOR--PSS(R)E																	SUN, OCT 06 2013 16:48	
IEEE BENCHMARK SYSTEM																	%MVA FOR TRANSFORMERS	
BRAZILIAN 7-BUS EQUIVALENT SYSTEM																	%MVA FOR NON-TRANSFORMER BRANCHES	
X-----	FROM BUS	-----X	AREA	VOLT	GEN	LOAD	SHUNT	X-----	TO BUS	-----X	AREA	CKT	MW	MVAR	TRANSFORMER			
BUS#	X--	NAME	--X	BASKV	ZONE	PU/KV	ANGLE	MW/MVAR	MM/MVAR	MM/MVAR	MM/MVAR				RATIO	ANGLE		
1	FOZ	AREIA	500.00	1	1.0300	24.5	1658.0	2405.0	0.0	0.0	3 S. SEGRED0	500.00	1	1	-999.9	124.3		
				1	515.00		-412.3R	-467.0	-190.1		5 IVAIPORA	500.00	1	1	252.9	120.5		
2	S.	SANTIAGO	500.00	1	1.0300	27.2	1332.0	692.3	0.0	0.0	3 S. SEGRED0	500.00	1	1	151.1	4.4		
				1	515.00		-200.3R	-184.0	-158.2		5 IVAIPORA	500.00	1	1	488.6	137.5		
3	S.	SEGRED0	500.00	1	1.0290	26.6	1540.0	688.2	0.0	0.0	1 FOZ AREIA	500.00	1	1	1002.8	-88.0		
				1	514.50		-446.7R	-235.0	-120.9		2 S. SANTIAGO	500.00	1	1	-151.0	-2.8		
4	ITAIPU	765.00	2	1.0390	48.5	6500.0	62.6	0.0	0.0	0.0	6 IVAIPORA	765.00	2	1	6437.4	1973.4		
				1	794.84		1958.0R	24.3	-39.7									
5	IVAIPORA	500.00	1	0.9984	21.2	0.0	845.8	0.0	0.0	0.0	1 FOZ AREIA	500.00	1	1	-251.5	-102.3		
				1	499.18		0.0	-9.2	-32.9		2 S. SANTIAGO	500.00	1	1	-484.9	-82.8		
											6 IVAIPORA	765.00	2	1	-109.3	227.3		
6	IVAIPORA	765.00	2	0.9895	21.4	0.0	0.0	-4.9	0.0	0.0	4 ITAIPU	765.00	2	1	-6315.6	1109.0		
				1	756.96		0.0	79.8	-2097.2		5 IVAIPORA	500.00	1	1	109.3	-224.8		
											7 EQUIVALENT	765.00	3	1	6211.2	1133.2		
7	EQUIVALENT	765.00	3	0.9660	0.0	-3164.3	2884.0	0.0	0.0	0.0	6 IVAIPORA	765.00	2	1	-6048.3	1187.5		
				1	738.99		952.4R	-196.0	-39.2									

3. Dynamic Simulation Models

The models and associated parameters for the dynamic simulation models used in this PSS[®]E setup are described in this Section. All generation units are represented by the salient-pole generator model GENSAE (5th order model). The excitation systems are represented by the simplified model SEXS, and the power system stabilizers are modeled by the speed-based single-input model IEEEEST.

3.1 Synchronous Machines

The generator model to represent the salient pole units is the PSS[®]E model GENSAE, shown in the block diagram in Figure 2. Details about the implementation of the model are available in the software documentation [2]. This is a 5th order dynamic model with the saturation function represented as a geometric (exponential) function.

The details associated with the representation of the saturation of the generators should not dramatically interfere with the results of a small-signal (linearized) analysis of the system performance. On the other hand, the proper representation of saturation is extremely important for transient stability and the determination of rated and ceiling conditions (minimum and maximum generator field current and generator field voltage) for the excitation system. The saturation factors S(1.0) and S(1.2) shown in Table 8 essentially neglect saturation effects.

The parameters for the model are presented in Table 8. Speed damping D is set to zero for all generators, as should be the case when using sub-transient generator models. Table 8 also presents the calculated rated field current (considering 0.80 rated power factor). This calculation comprises the initialization of the generators models at full (rated) power output, considering their rated power factor. It should be noted that in PSS[®]E models, due to the choice of base values for generator field voltage and generator field current, these variables are numerically the same, in steady state, when expressed in per unit (pu).

Table 8: Dynamic Model Data for Salient Pole Units (PSS[®]E Model GENSAE)

Parameters			Buses	
Description	Symbol	Unit	1 FOZ AREIA	2 S. SANTIAGO
Rated apparent power	MBASE	MVA	1900	1400
d-axis open circuit transient time constant	T'_{do}	s	5.0	5.0
d-axis open circuit sub-transient time constant	T''_{do}	s	0.053	0.053
q-axis open circuit sub-transient time constant	T''_{qo}	s	0.123	0.123
Inertia	H	MW.s/MVA	4.5	4.5
Speed damping	D	pu	0.0	0.0
d-axis synchronous reactance	X_d	pu	0.85	0.85
q-axis synchronous reactance	X_q	pu	0.7	0.7
d-axis transient reactance	X'_d	pu	0.3	0.3
sub-transient reactance	$X''_d = X''_q$	pu	0.2	0.2
Leakage reactance	X_ℓ	pu	0.15	0.15
Saturation factor at 1.0 pu voltage	S(1.0)	–	0.001	0.001
Saturation factor at 1.2 pu voltage	S(1.2)	–	0.01	0.01
Rated field current	IFD _{rated}	pu	1.66	1.66

Table 8 (cont.): Dynamic Model Data for Salient Pole Units (PSS®E Model GENSAE)

Parameters		Buses		
Symbol	Unit	3 S. SEGREDO	4 ITAIPU	7 EQUIVALENT
MBASE	MVA	1944	6633	6000
T'_{do}	s	5.0	7.6	8.0
T''_{do}	s	0.06	0.09	0.09
T''_{qo}	s	0.09	0.19	0.2
H	MW.s/MVA	4.5	5.07	5.0
D	pu	0.0	0.0	0.0
X_d	pu	0.88	0.9	1.0
X_q	pu	0.69	0.68	0.7
X'_d	pu	0.3	0.3	0.3
$X''_d = X''_q$	pu	0.2	0.24	0.25
X_t	pu	0.15	0.18	0.18
S(1.0)	–	0.001	0.001	0.001
S(1.2)	–	0.01	0.01	0.01
IFD _{rated}	pu	1.68	1.70	1.79

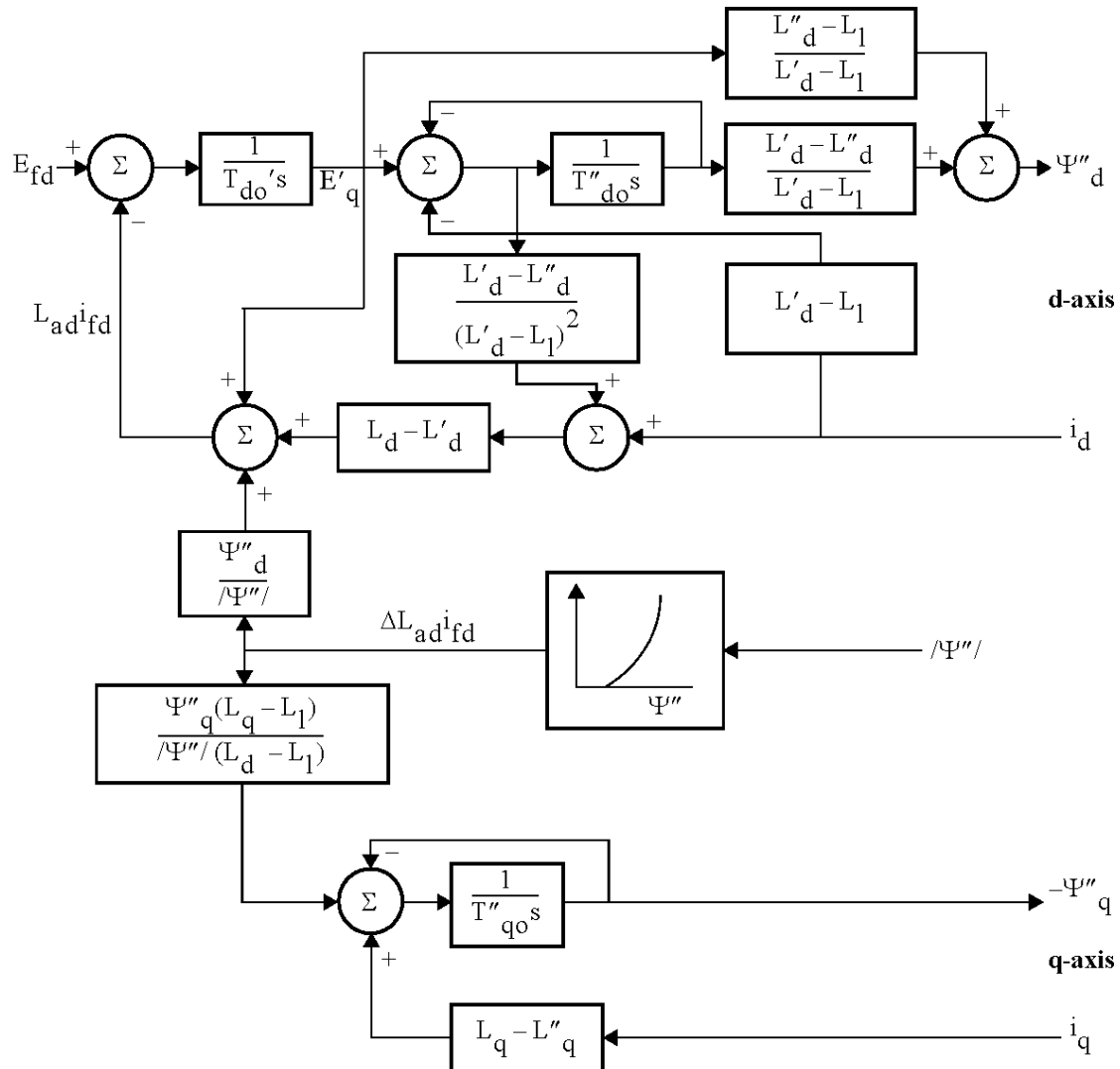


Figure 2: Block Diagram for the Salient Pole Generator (PSS®E Model GENSAE)

3.2 Excitation Systems

All excitation systems are considered identical and will be represented by the same dynamic model and parameters. The PSS[®]E simplified excitation system model SEXS [2], shown in Figure 3, is used. The parameters for the model are presented in Table 9. The limits $E_{\max}$ and $E_{\min}$ are set to reasonably large values, so they should not interfere and affect small-signal stability results.

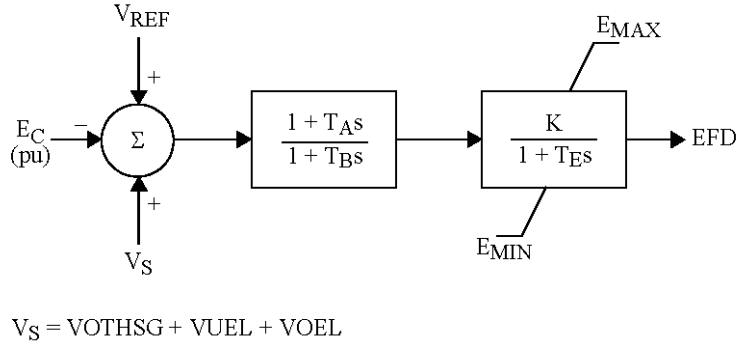


Figure 3: Block Diagram for the PSS[®]E Model SEXS

Table 9: Dynamic Model Data for the Simple Excitation System (PSS[®]E Model SEXS)

Parameters			
Description	Symbol	Value	Unit
TGR block 1 transient gain	TA/TB	1	–
TGR block 1 denominator time constant	TB	1	s
Exciter gain	K	30	pu
Exciter time constant	TE	0.05	s
Max. AVR output	Emin	–4	pu
Min. AVR output	Emax	5	pu

3.3 Power System Stabilizers

This Benchmark system considers power system stabilizers (PSS) in all machines except the machine connected to bus #7 (EQUIVALENT). All PSSs are derived from rotor speed deviation and have the same structure, two lead-lag blocks, one wash-out block, and a gain.

The IEEE Std. 421.5(2005) model PSS1A [5] will be used to represent the power system stabilizers. This model can be represented in PSS[®]E by the IEEEEST model [2]. The block diagram of the IEEEEST model is shown in Figure 4 and the parameters are given in Table 10. It should be noted that the parameters A1 to A6 are set in a way that leads to the whole filter block to be bypassed.

The output limits were set to $\pm 10\%$, while the logic to switch off the PSS for voltages outside a normal operation range has been ignored (parameters V_{CU} and V_{CL} set to zero). The input signal is rotor speed deviation.

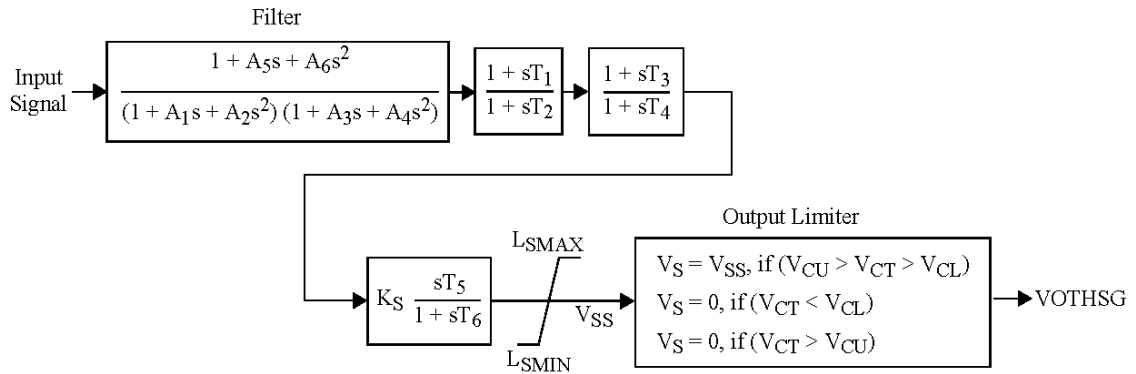


Figure 4: Block Diagram for the PSS®E Model IEEEEST

Table 10: Dynamic Model Data for Power System Stabilizers (PSS®E Model IEEEEST)

Parameters			Buses	
Description	Symbol	Unit	1, 2, 3	4
2 nd order denominator coefficient	A ₁	—	0	0
2 nd order denominator coefficient	A ₂	—	0	0
2 nd order numerator coefficient	A ₃	—	0	0
2 nd order numerator coefficient	A ₄	—	0	0
2 nd order denominator coefficient	A ₅	—	0	0
2 nd order denominator coefficient	A ₆	—	0	0
1 st lead-lag numerator time constant	T ₁	s	0.30	0.52
1 st lead-lag denominator time constant	T ₂	s	0.075	0.065
2 nd lead-lag numerator time constant	T ₃	s	0.30	0.52
2 nd lead-lag denominator time constant	T ₄	s	0.075	0.065
Washout block numerator time constant	T ₅	s	3	3
Washout block denominator time constant	T ₆	s	3	3
PSS gain	K _S	pu	10	16
PSS max. output	L _{Smax}	pu	0.1	0.1
PSS min. output	L _{Smin}	pu	-0.1	-0.1
Upper voltage limit for PSS operation	V _{CU}	pu	0	0
Lower voltage limit for PSS operation	V _{CL}	pu	0	0

4. Simulation Results

The results presented in this report correspond to time-domain simulations of different disturbances. The main objective associated with the selection of these disturbances was to assess the system damping and the effectiveness of the proposed stabilizers in providing damping to these oscillations.

This benchmark system, as described, is oscillatory unstable without power system stabilizers. All linear techniques for locating stabilizers (e.g., transfer function residues, participation factors, etc.) indicate that the unstable mode can be stabilized by a PSS at Itaipu (bus #4).

But the key characteristic of this system is the presence of a zero in the transfer function associated with the PSS at Itaipu (bus #4) that precludes proper damping of the mode with just one PSS at Itaipu [1]. Thus, stabilizers at the other machines (except the equivalent machine at bus #7) are necessary to eliminate this poorly located transfer function zero and thus achieve sufficient damping for the unstable oscillation mode.

The first set of simulations (Figure 5 to Figure 9) comprise the connection of a 50 Mvar reactor at the Ivaiporã 765 kV bus (bus #6) at $t=1.0$ second. The reactor is disconnected 100 ms later, without any changes in the system topology. This is a very small disturbance, and as such leads to results that correspond to the linear response of the system.

Figure 5 presents the electrical power output of the Itaipu machine (bus #4), in MW. The black trace corresponds to the case without any PSS in service, and the case is unstable (negative damping). The blue trace is the result with only the Itaipu PSS in service. The case is no longer unstable, but the damping is clearly insufficient. The red trace shows the results with PSSs in service at all machines but the equivalent generator (bus #7) and it can be seen that the oscillation is now properly damped.

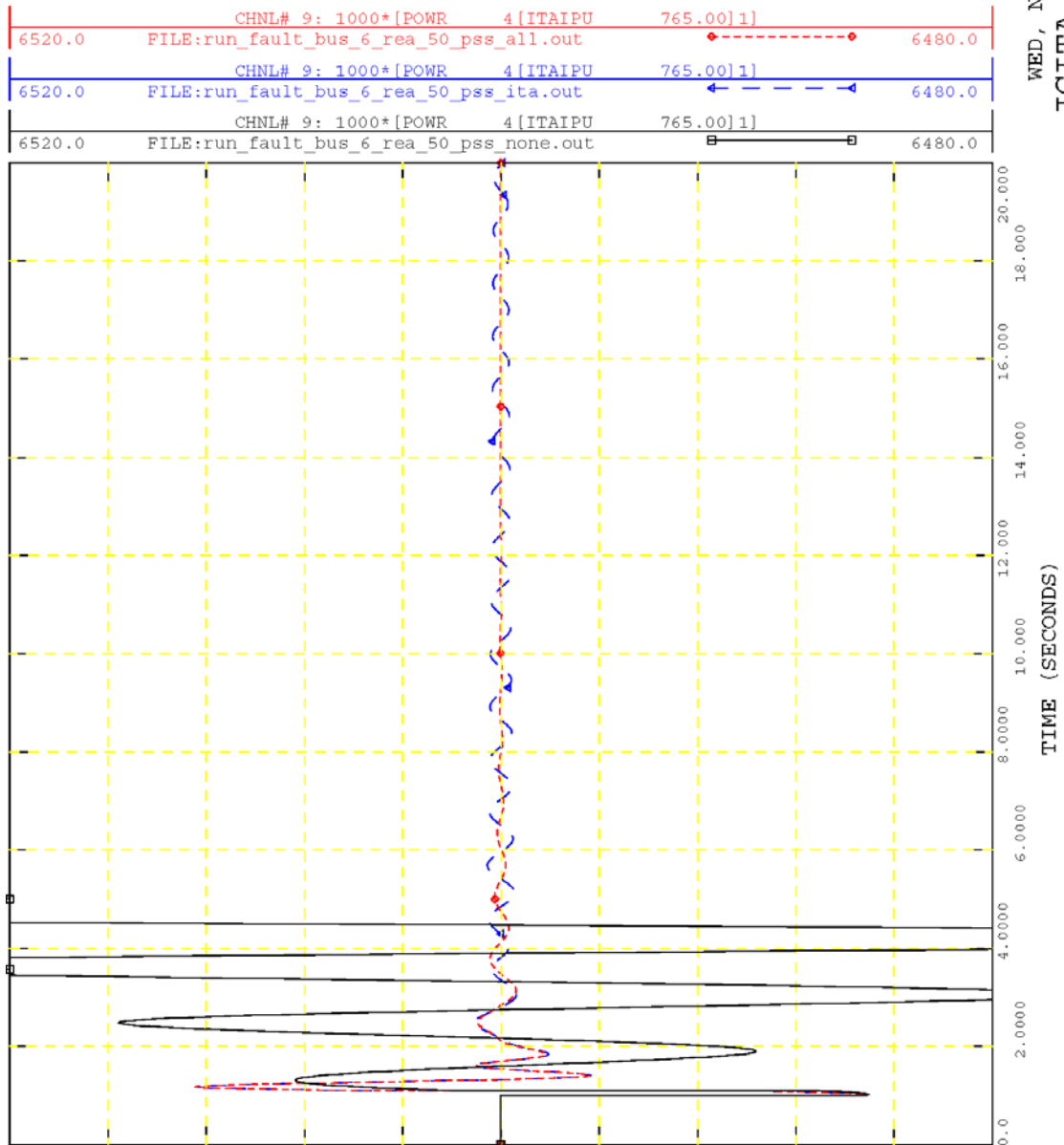
The electrical power outputs of the other units in the system are shown in Figure 6 to Figure 9.

The second set of simulations correspond to a 2% step in voltage reference at the Itaipu machine (bus #4), applied at $t=1.0$ second. The step is removed (−2% step) at $t=11.0$ seconds. Figure 10 to Figure 14 present the electrical power output of the generation units, for the cases without PSS (black trace), with only the Itaipu PSS (blue trace) and with PSSs in all machines except the equivalent generator at bus #7 (red trace).

4.1 50 Mvar Reactor at Ivaiporã Bus



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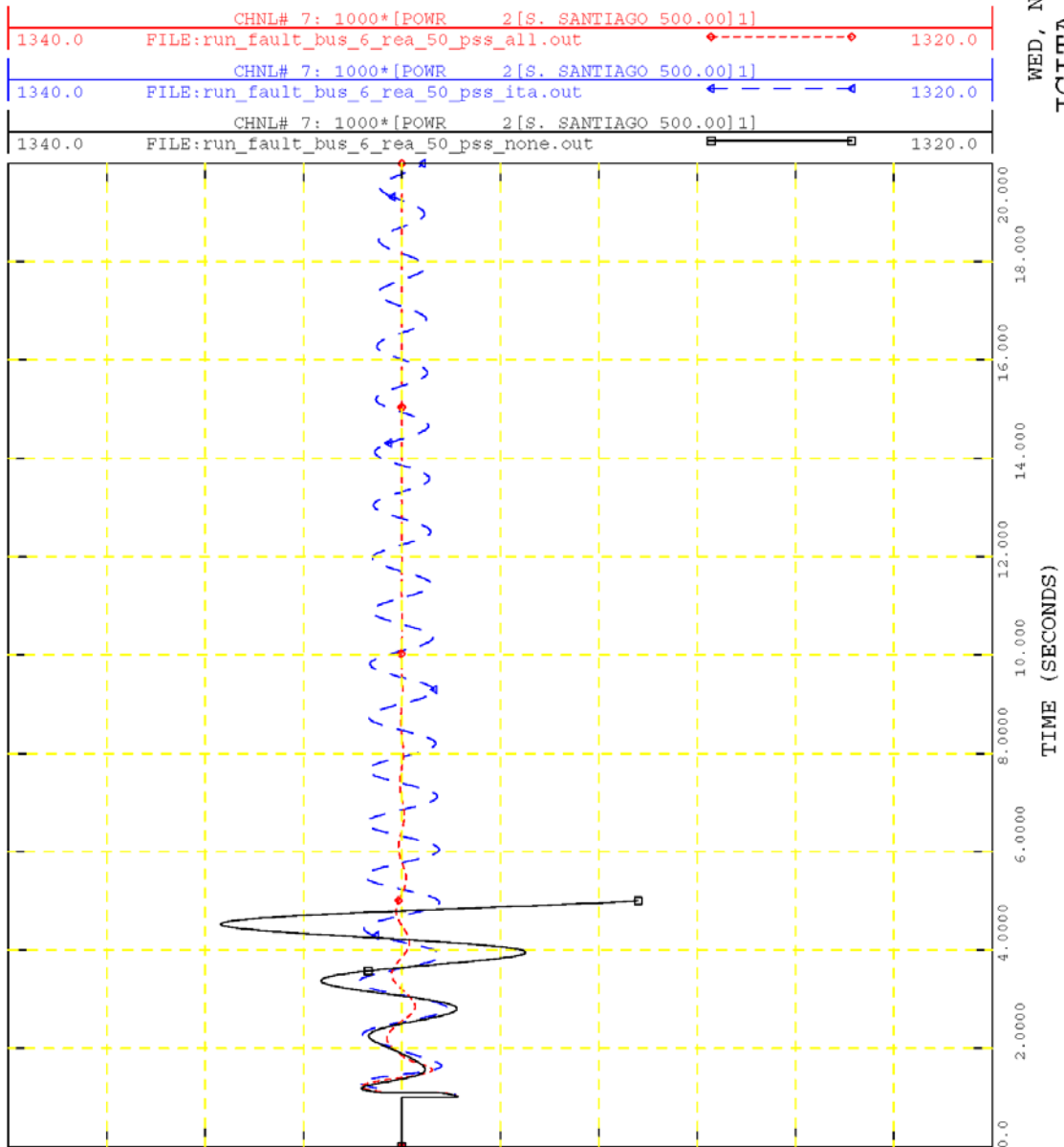


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Figure 5: Itaipú Unit (Bus #4) Electrical Power Output in MW



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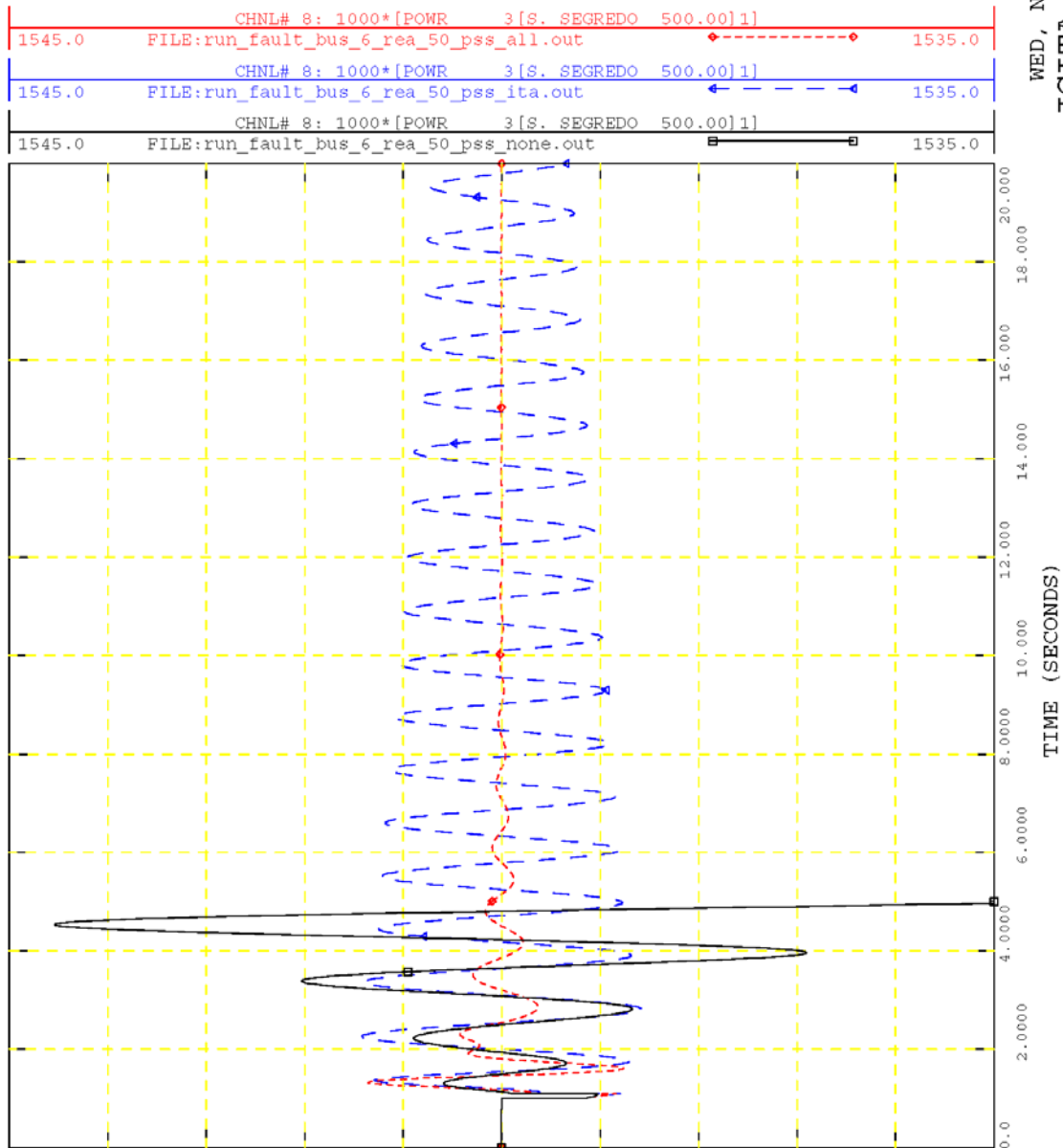


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Figure 6: Salto Santiago (Bus #2) Unit Electrical Power Output in MW



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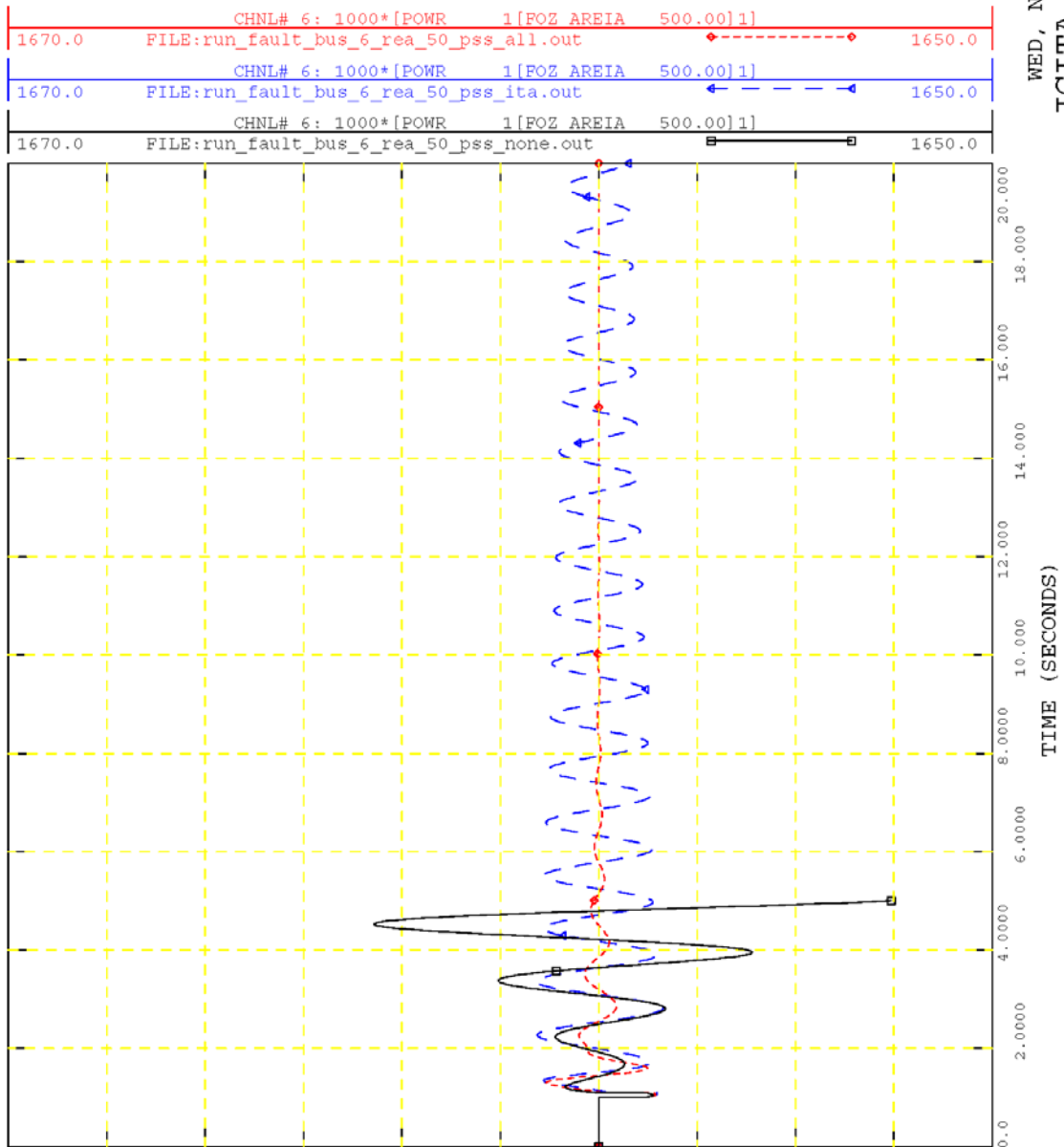


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Figure 7: Salto Segredo Unit (Bus #3) Electrical Power Output in MW



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Figure 8: Foz do Areia Unit (Bus #1) Electrical Power Output in MW



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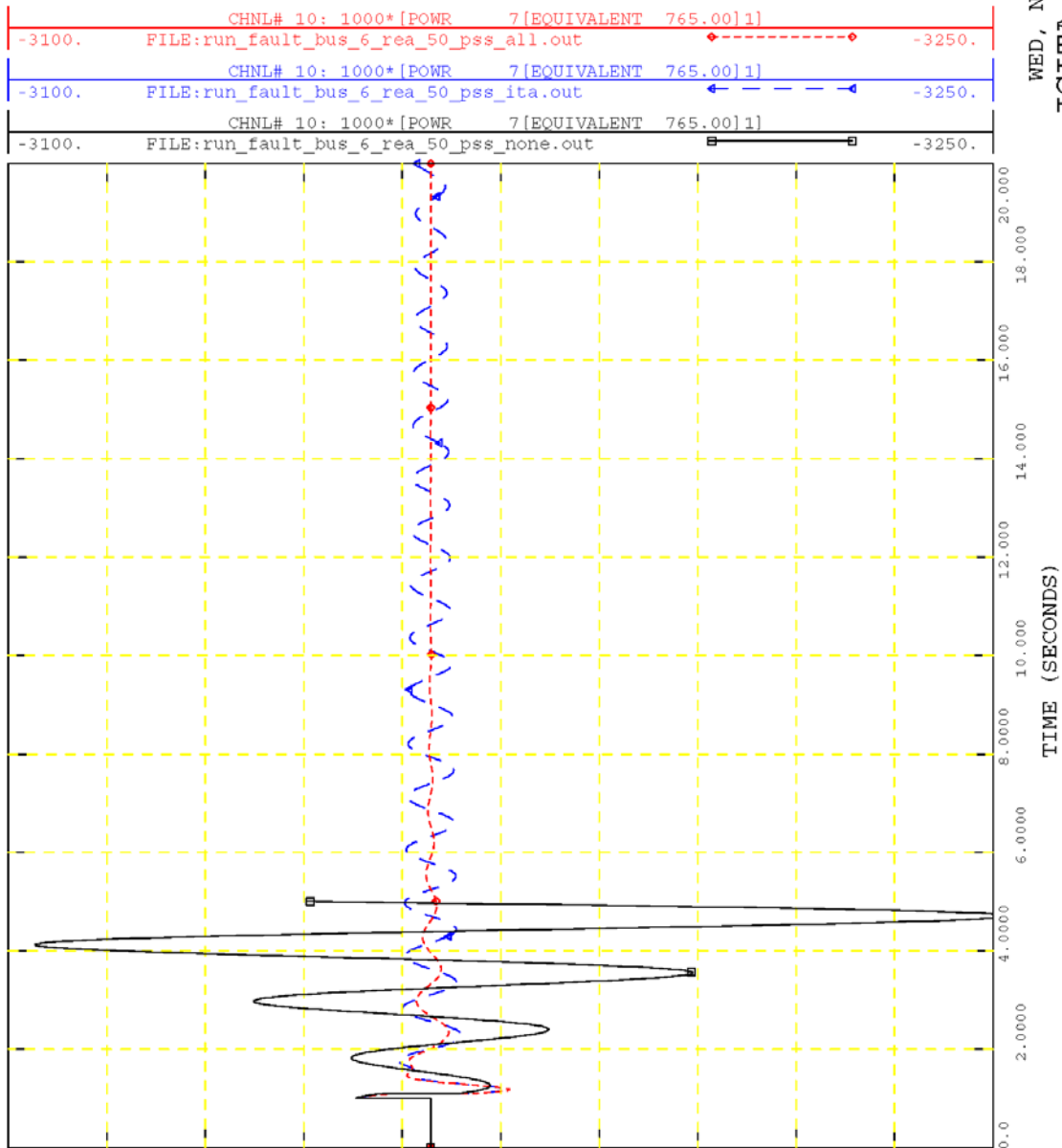


Figure 9: Equivalent Unit (Bus #7) Electrical Power Output in MW

4.2 Step at Itaipu Voltage Reference



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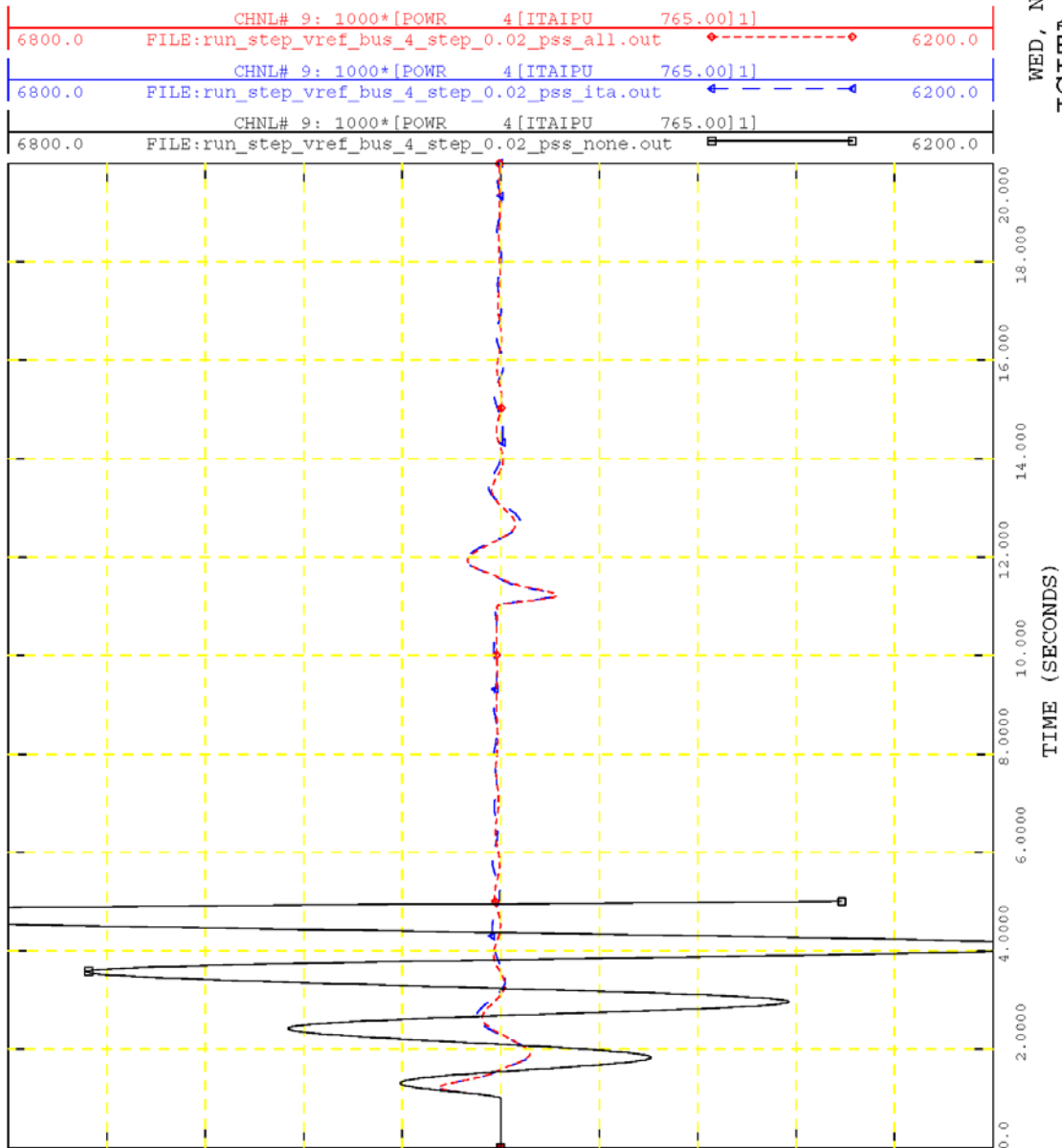


Figure 10: Itaipu Unit (Bus #4) Electrical Power Output in MW



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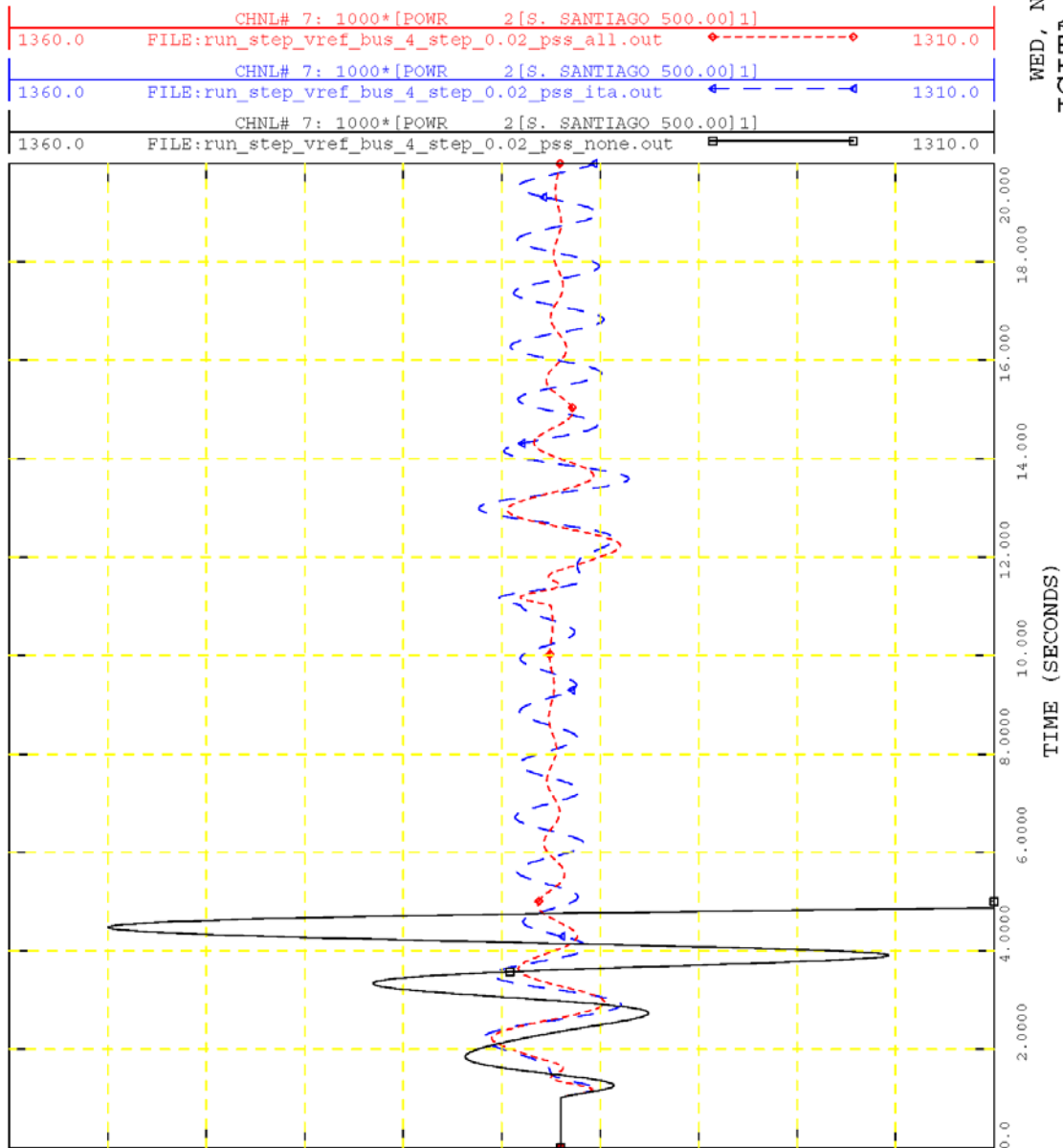


Figure 11: Salto Santiago Unit (Bus #2) Electrical Power Output in MW



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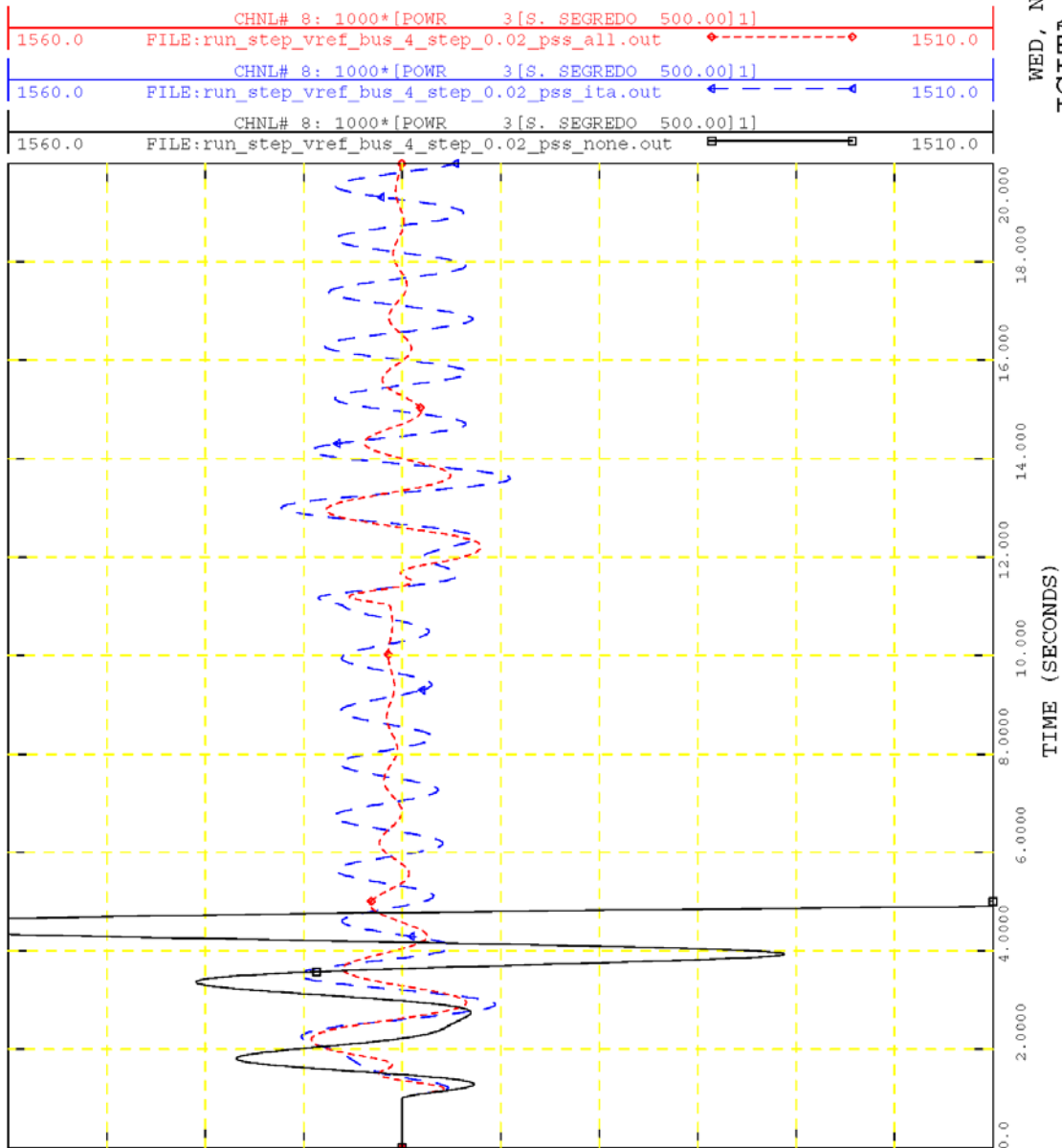


Figure 12: Salto Segredo Unit (Bus #3) Electrical Power Output in MW



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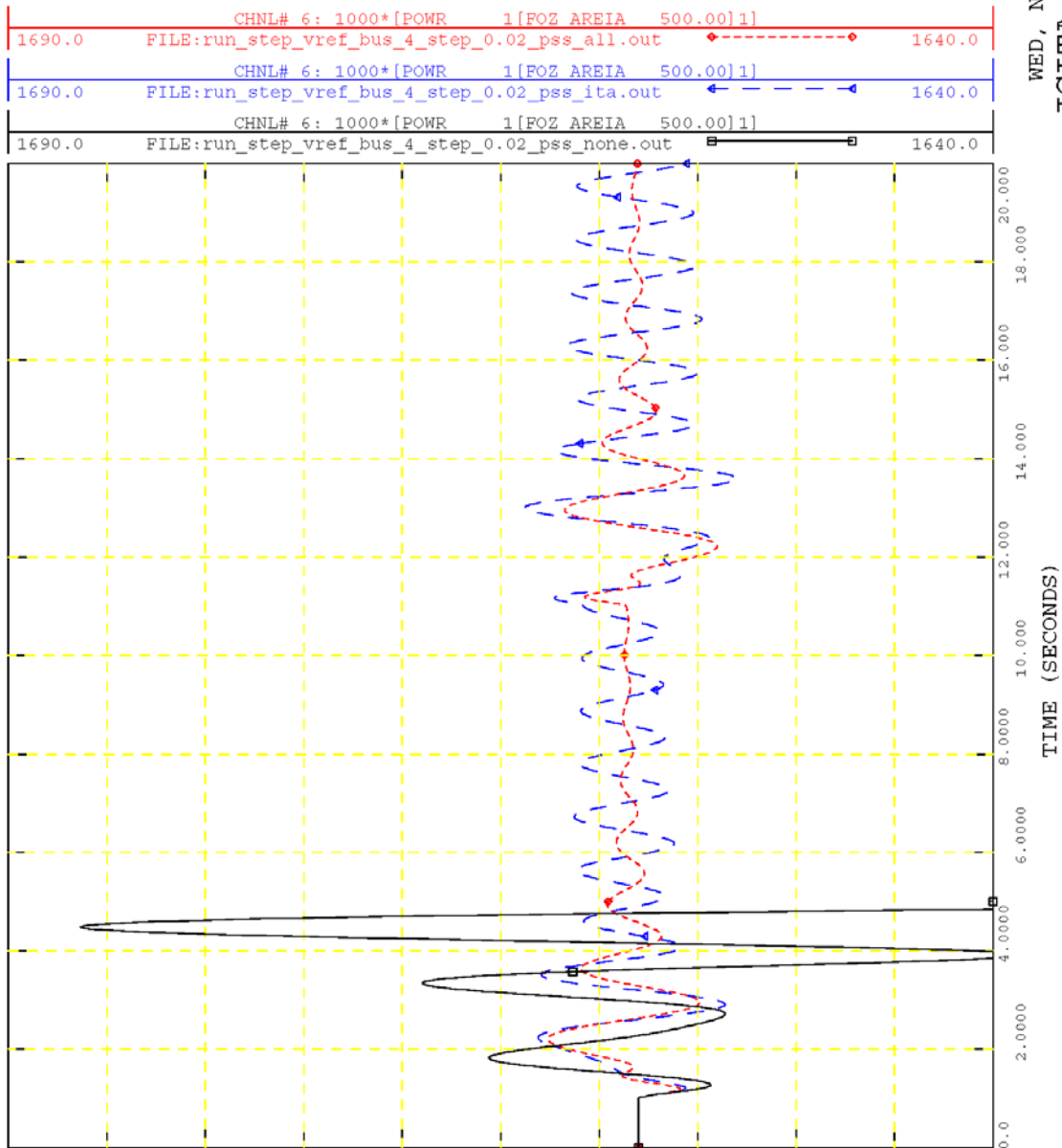


Figure 13: Foz do Areia Unit (Bus #1) Electrical Power Output in MW



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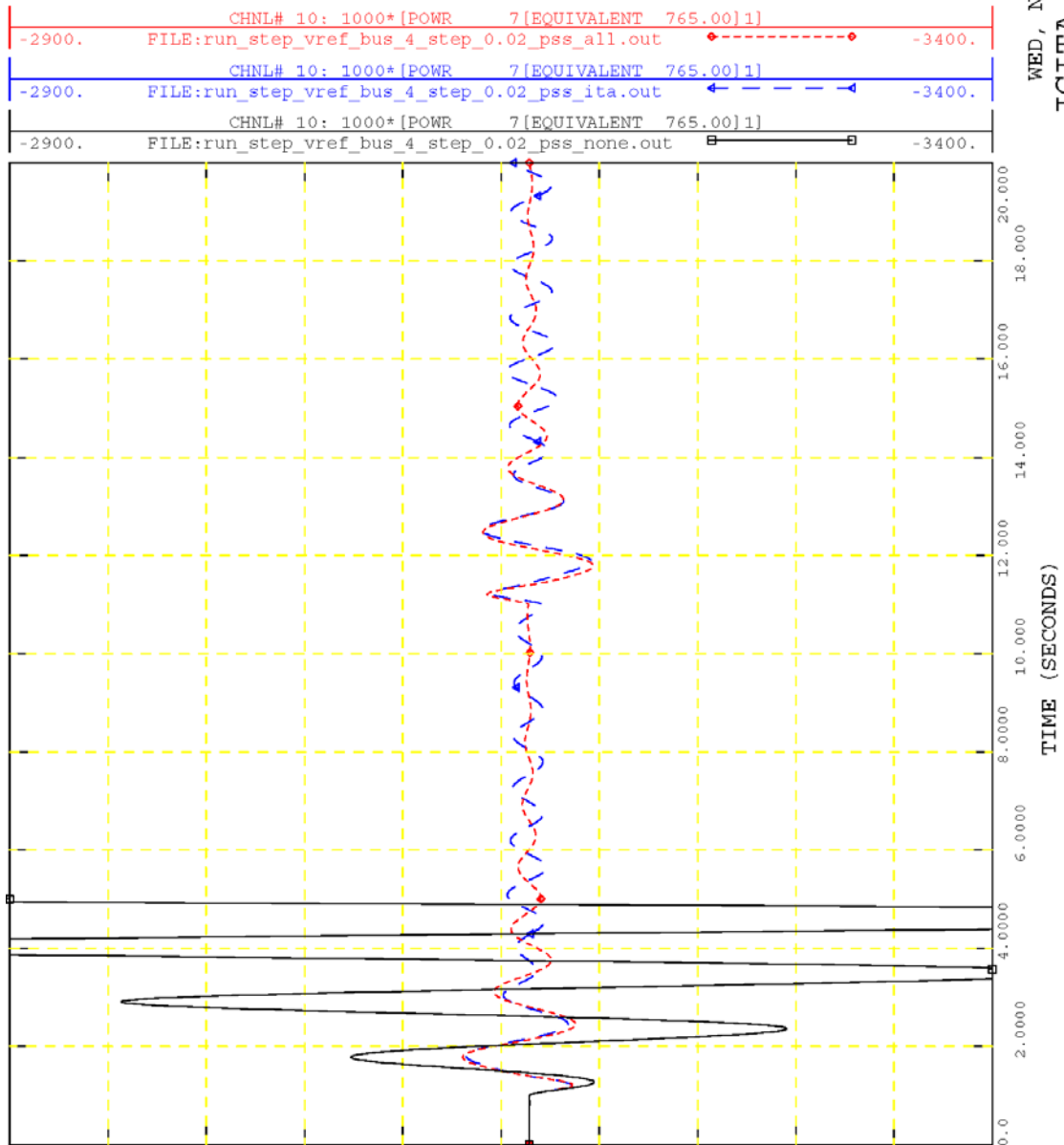


Figure 14: Equivalent Unit (Bus #7) Electrical Power Output in MW

4.3 Modal Analysis

PSS®E auxiliary program PSSPLT¹ can estimate the dominant modes from one system by decomposing its oscillatory response. The following tables show the results using the Least Square decomposition method.

4.3.1 No PSS

Table 11 shows the “original” JCITA system eigenvalues. It can be seen that there is a complex pair with the real part positive which characterizes an unstable mode.

Table 11: Eigenvalues Calculated with PSS®E Program PSSPLT

PTI INTERACTIVE POWER SYSTEM SIMULATOR--PSS/E

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IEEE BENCHMARK SYSTEM

BRAZILIAN 7-BUS EQUIVALENT SYSTEM

EIGENVALUES:

NO.	REAL	IMAG	DAMP	FREQ
1	-1.8470	9.1770	0.19731	1.4606
2	-1.8470	-9.1770	0.19731	1.4606
3	-2.0556	9.1654	0.21885	1.4587
4	-2.0556	-9.1654	0.21885	1.4587
5	-0.29081	5.8667	0.49508E-01	0.93371
6	-0.29081	-5.8667	0.49508E-01	0.93371
7	-6.4142	5.5609	0.75557	0.88505
8	-6.4142	-5.5609	0.75557	0.88505
9	0.69914	5.4366	-0.12755	0.86526
10	0.69914	-5.4366	-0.12755	0.86526
11	-6.6039	2.1971	0.94887	0.34967
12	-6.6039	-2.1971	0.94887	0.34967
13	-7.7453	1.1572	0.98902	0.18418
14	-7.7453	-1.1572	0.98902	0.18418
15	-18.587	0.80059	0.99907	0.12742
16	-18.587	-0.80059	0.99907	0.12742
17	-0.13523E-01	0.62574E-01	0.21123	0.99589E-02
18	-0.13523E-01	-0.62574E-01	0.21123	0.99589E-02

¹ Provided as part of the PSS®E installation, for users with the proper license. Consult the software vendor if you are not sure about your license.

CHANNEL FILE: run_fault_bus_6_rea_50_pss_none.out

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BRAZILIAN 7-BUS EQUIVALENT SYSTEM

CHANNEL: CHNL# 34: [SPD 4[ITAIPU 765.00]1]
TIME INTERVAL: 1.2000 - 4.7000 SEC.

MODAL COMPONENTS

COMP. NO	EIGENVALUE		EIGENVECTOR		REMARKS
	REAL	IMAGINARY	MAGNITUDE	ANGLE	
1	0.707567	5.44299	0.16822E-04	67.30	FREQ.: 0.866 HZ.
2	-2.46460	--	0.44952E-05	--	TCNST: 0.406 SC.
3	-34.5179	182.730	0.66728E-05	43.62	FREQ.: 29.082 HZ.
4	-9.14982	55.7144	0.36196E-05	-98.57	FREQ.: 8.867 HZ.
5	-12.0880	114.826	0.26862E-05	-68.32	FREQ.: 18.275 HZ.
6	-11.0529	138.551	0.20842E-05	-28.50	FREQ.: 22.051 HZ.
7	-8.72382	85.4886	0.20348E-05	-67.37	FREQ.: 13.606 HZ.
8	-0.234707	10.7970	0.19590E-05	-80.18	FREQ.: 1.718 HZ.
9	-9.87398	175.272	0.14216E-05	-88.79	FREQ.: 27.895 HZ.

PERC. ERROR: 1.922
SIGNAL/NOISE: 34.70
METHODS: EIGENVALUE - LEAST SQUARE, EIGENVECTOR - LEAST SQUARE
ORDER: 30

CHANNEL FILE: run_fault_bus_6_rea_50_pss_none.out

IEEE BENCHMARK SYSTEM
BRAZILIAN 7-BUS EQUIVALENT SYSTEM

CHANNEL: CHNL# 32: [SPD 2[S. SANTIAGO 500.00]1]
TIME INTERVAL: 1.2000 - 4.0000 SEC.

MODAL COMPONENTS

COMP. NO	EIGENVALUE		EIGENVECTOR		REMARKS
	REAL	IMAGINARY	MAGNITUDE	ANGLE	
1	-2.71953	--	0.49004E-05	--	TCNST: 0.368 SC.
2	0.739907	5.48148	0.46626E-05	141.87	FREQ.: 0.872 HZ.
3	-3.72779	10.1187	0.42710E-05	71.22	FREQ.: 1.610 HZ.
4	-10.2605	61.3625	0.21551E-06	98.22	FREQ.: 9.766 HZ.
5	-10.4198	121.613	0.12378E-06	155.86	FREQ.: 19.355 HZ.
6	-7.84055	95.6032	0.11294E-06	116.46	FREQ.: 15.216 HZ.
7	-8.23933	155.066	0.68604E-07	-179.76	FREQ.: 24.680 HZ.
8	-7.56744	186.549	0.47514E-07	-148.40	FREQ.: 29.690 HZ.

PERC. ERROR: 3.640
SIGNAL/NOISE: 27.10
METHODS: EIGENVALUE - LEAST SQUARE, EIGENVECTOR - LEAST SQUARE
ORDER: 27

4.3.2 PSS at Itaipu

NUMBER OF SELECTED DATA POINTS IS 860 AND TIME STEP IS 0.008
 SELECT AN NPLT OF 8 TO REDUCE THE NUMBER OF POINTS TO ABOUT 100
 EFFECTIVE NUMBER OF DATA POINTS IS 286 AND EFFECTIVE TIME STEP IS 0.025

CHANNEL FILE: run_fault_bus_6_rea_50_pss_ita.out

IEEE BENCHMARK SYSTEM
 BRAZILIAN 7-BUS EQUIVALENT SYSTEM

CHANNEL: CHNL# 34: [SPD 4[ITAIPU 765.00]1]
 TIME INTERVAL: 1.7833 - 8.9416 SEC.

MODAL COMPONENTS

COMP. NO	EIGENVALUE		EIGENVECTOR		REMARKS
	REAL	IMAGINARY	MAGNITUDE	ANGLE	
1	-0.700954	3.88240	0.31665E-05	-90.53	FREQ.: 0.618 HZ.
2	-0.392241E-01	5.83679	0.14629E-05	-134.37	FREQ.: 0.929 HZ.
3	-0.600196E-01	--	0.58355E-06	--	TCNST: 16.661 SC.
4	-3.96201	14.1973	0.93869E-06	-172.52	FREQ.: 2.260 HZ.
5	-132.750	--	0.18005E-07	--	TCNST: 0.008 SC.
6	-6.11962	51.7513	0.56811E-08	-56.20	FREQ.: 8.236 HZ.
7	-2.07211	17.2938	0.54852E-08	-34.19	FREQ.: 2.752 HZ.
8	-3.02469	32.2951	0.33751E-08	-61.61	FREQ.: 5.140 HZ.
9	-1.73617	26.4327	0.97554E-09	-105.83	FREQ.: 4.207 HZ.
10	-1.65890	39.4505	0.91572E-09	-63.99	FREQ.: 6.279 HZ.
11	-1.79492	48.1278	0.90750E-09	-163.90	FREQ.: 7.660 HZ.
12	-1.61872	57.8029	0.82658E-09	41.91	FREQ.: 9.200 HZ.

PERC. ERROR: 1.566
 SIGNAL/NOISE: 36.72
 METHODS: EIGENVALUE - LEAST SQUARE, EIGENVECTOR - LEAST SQUARE
 ORDER: 42

NUMBER OF SELECTED DATA POINTS IS 967 AND TIME STEP IS 0.008
 SELECT AN NPLT OF 9 TO REDUCE THE NUMBER OF POINTS TO ABOUT 100
 EFFECTIVE NUMBER OF DATA POINTS IS 322 AND EFFECTIVE TIME STEP IS 0.025

CHANNEL FILE: run_fault_bus_6_rea_50_pss_ita.out

IEEE BENCHMARK SYSTEM
 BRAZILIAN 7-BUS EQUIVALENT SYSTEM

CHANNEL: CHNL# 32: [SPD 2[S. SANTIAGO 500.00]1]
 TIME INTERVAL: 1.2833 - 9.3333 SEC.

MODAL COMPONENTS

COMP. NO	EIGENVALUE		EIGENVECTOR		REMARKS
	REAL	IMAGINARY	MAGNITUDE	ANGLE	
1	-0.417610E-01	5.83244	0.12416E-04	119.33	FREQ.: 0.928 HZ.
2	-1.06764	3.85172	0.73100E-05	-41.44	FREQ.: 0.613 HZ.
3	-1.94027	9.42923	0.37704E-05	118.90	FREQ.: 1.501 HZ.
4	-4.06151	14.2612	0.24291E-05	-38.81	FREQ.: 2.270 HZ.
5	-0.154238	--	0.85108E-06	--	TCNST: 6.483 SC.
6	-3.26438	44.6039	0.59773E-07	24.10	FREQ.: 7.099 HZ.
7	-2.56384	42.0519	0.44050E-07	-89.75	FREQ.: 6.693 HZ.
8	-2.17107	32.4173	0.23673E-07	-47.05	FREQ.: 5.159 HZ.
9	-2.24696	52.2368	0.18447E-07	34.81	FREQ.: 8.314 HZ.
10	-2.15622	57.1731	0.13831E-07	90.55	FREQ.: 9.099 HZ.
11	-0.877513	25.2768	0.10091E-07	-31.63	FREQ.: 4.023 HZ.

PERC. ERROR: 1.582
 SIGNAL/NOISE: 34.92
 METHODS: EIGENVALUE - LEAST SQUARE, EIGENVECTOR - LEAST SQUARE
 ORDER: 41

4.3.3 PSS at All Machines

NUMBER OF SELECTED DATA POINTS IS 693 AND TIME STEP IS 0.008					
SELECT AN NPLT OF 6 TO REDUCE THE NUMBER OF POINTS TO ABOUT 100					
EFFECTIVE NUMBER OF DATA POINTS IS 231 AND EFFECTIVE TIME STEP IS 0.025					
CHANNEL FILE: run_fault_bus_6_rea_50_pss_all.out					
IEEE BENCHMARK SYSTEM					
BRAZILIAN 7-BUS EQUIVALENT SYSTEM					
CHANNEL: CHNL# 34: [SPD 4[ITAIPU 765.00]1]					
TIME INTERVAL: 1.8250 - 7.5916 SEC.					
MODAL COMPONENTS					
COMP. NO	EIGENVALUE		EIGENVECTOR		REMARKS
	REAL	IMAGINARY	MAGNITUDE	ANGLE	
1	-0.437235	4.88936	0.31565E-05	-104.73	FREQ.: 0.778 HZ.
2	-1.29290	3.41596	0.12155E-05	-52.41	FREQ.: 0.544 HZ.
3	-4.29502	14.2399	0.80381E-06	-143.46	FREQ.: 2.266 HZ.
4	-0.486911E-01	--	0.35360E-06	--	TCNST: 20.538 SC.
5	-2.05888	17.9331	0.45370E-08	119.05	FREQ.: 2.854 HZ.
6	-2.73865	23.1637	0.16747E-08	172.61	FREQ.: 3.687 HZ.
7	-2.09055	37.2507	0.12660E-08	169.77	FREQ.: 5.929 HZ.
8	-2.01404	56.5822	0.75808E-09	-154.29	FREQ.: 9.005 HZ.
9	-1.00134	43.9675	0.43750E-09	-167.74	FREQ.: 6.998 HZ.
10	-1.05279	49.7804	0.38606E-09	-159.99	FREQ.: 7.923 HZ.
11	-0.741519	31.0985	0.29546E-09	114.03	FREQ.: 4.949 HZ.
PERC. ERROR: 0.8675					
SIGNAL/NOISE: 41.29					
METHODS: EIGENVALUE - LEAST SQUARE, EIGENVECTOR - LEAST SQUARE					
ORDER: 43					

NUMBER OF SELECTED DATA POINTS IS 783 AND TIME STEP IS 0.008					
SELECT AN NPLT OF 7 TO REDUCE THE NUMBER OF POINTS TO ABOUT 100					
EFFECTIVE NUMBER OF DATA POINTS IS 261 AND EFFECTIVE TIME STEP IS 0.025					
CHANNEL FILE: run_fault_bus_6_rea_50_pss_all.out					
IEEE BENCHMARK SYSTEM					
BRAZILIAN 7-BUS EQUIVALENT SYSTEM					
CHANNEL: CHNL# 32: [SPD 2[S. SANTIAGO 500.00]1]					
TIME INTERVAL: 1.6000 - 8.1166 SEC.					
MODAL COMPONENTS					
COMP. NO	EIGENVALUE		EIGENVECTOR		REMARKS
	REAL	IMAGINARY	MAGNITUDE	ANGLE	
1	-0.402555	4.83123	0.71027E-05	-85.37	FREQ.: 0.769 HZ.
2	-3.40692	--	-0.29306E-05	--	TCNST: 0.294 SC.
3	-14.2885	--	0.25267E-05	--	TCNST: 0.070 SC.
4	-3.64382	9.44457	0.26045E-05	-88.77	FREQ.: 1.503 HZ.
5	-0.182164	--	0.58536E-06	--	TCNST: 5.490 SC.
6	-2.08941	14.5850	0.36719E-06	-38.58	FREQ.: 2.321 HZ.
7	-3.32931	43.3634	0.75991E-07	-147.84	FREQ.: 6.902 HZ.
8	-2.86408	21.1938	0.69333E-07	-155.02	FREQ.: 3.373 HZ.
9	-3.60589	36.7381	0.69062E-07	175.95	FREQ.: 5.847 HZ.
10	-2.17182	46.6651	0.35010E-07	-58.67	FREQ.: 7.427 HZ.
11	-1.76623	29.7095	0.25797E-07	174.11	FREQ.: 4.728 HZ.
12	-1.97827	54.9647	0.15864E-07	-100.68	FREQ.: 8.748 HZ.
13	-2.15861	61.6569	0.13529E-07	-78.28	FREQ.: 9.813 HZ.
PERC. ERROR: 4.590					
SIGNAL/NOISE: 27.24					
METHODS: EIGENVALUE - LEAST SQUARE, EIGENVECTOR - LEAST SQUARE					
ORDER: 42					

5. References

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